



Remote Wireless Display Of Sensor Data

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Remote wireless display of sensor data is required in many industrial and consumer applications. Sensor data like light, temperature, accelerometer and strain gauge data, etc can be communicated through Wi-Fi and displayed on a mobile phone, laptop, or desktop PC

depending on application and convenience.

This project uses NodeMCU for this purpose, which is a single-board microcontroller board having Wi-Fi chip ESP8266. Hence it is good for IoT applications. NodeMCU has open source hardware and firmware that

uses Lua scripting language. It has following features:

- One analogue input (10-bit integrated ADC)
- Thirteen digital I/O pins (alternate functions like USART/I2C/SPI)
- Micro USB port for power, programming, and debugging
- A 2x2.54mm, 15-pin header with access to GPIOs, SPI, UART, ADC, and power pins
- Reset and flash buttons
- 5V power supply through the micro USB port
- Wi-Fi support (802.11 b/g/n, 2.4GHz, WPA/WPA2)
- Integrated low-power 32-bit RISC MCU (L106)
- 64KB instructions RAM, 96KB data RAM
- Integrated TCP/IP protocol stack
- Integrated TR switch, balun, LNA, power amplifier, and matching network
- Integrated PLL, regulators, and power management units
- Supports antenna diversity
- Standby power consumption of < 1.0mW
- + 20dBm output power in 802.11b mode
- Operating temperature range of -40°C to 125°C

The pin diagram of NodeMCU is

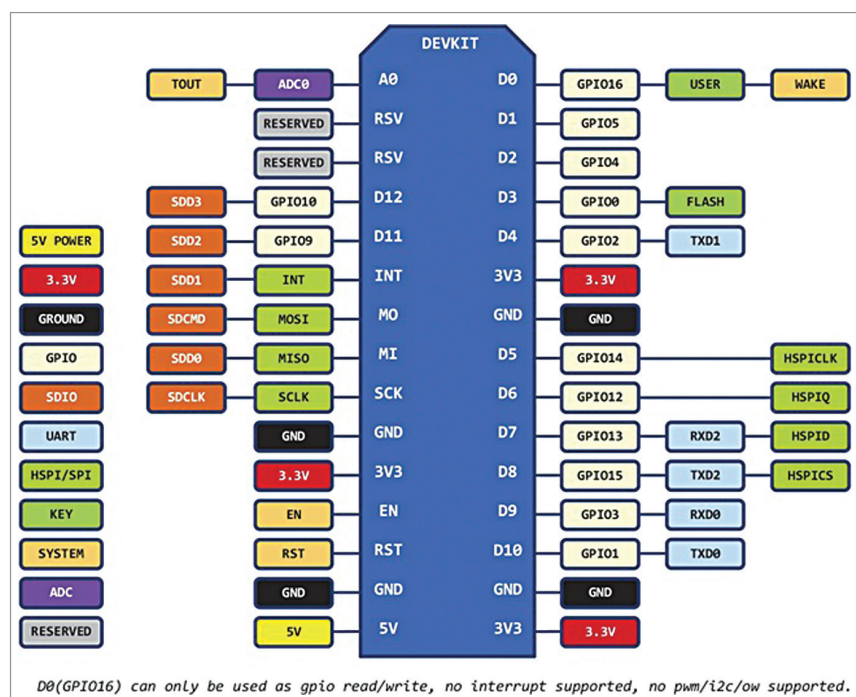


Fig. 1: Pin diagram of NodeMCU

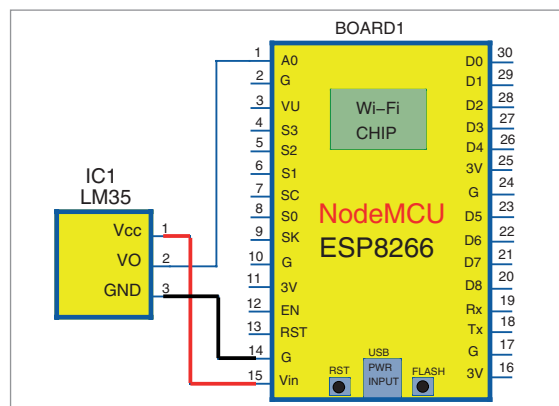


Fig. 2: Circuit diagram for interfacing LM35 sensor with NodeMCU

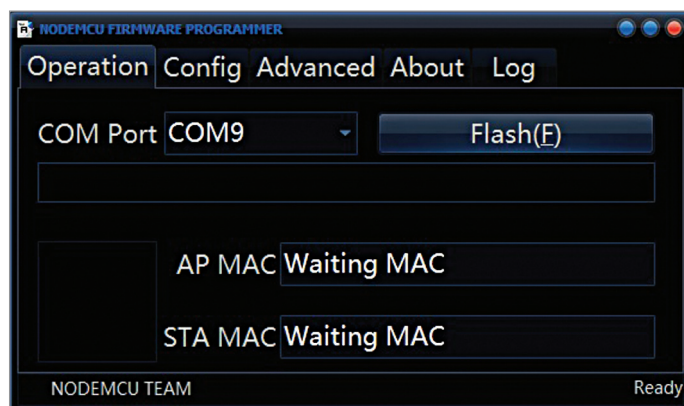


Fig. 3: Flash button in ESP8266 flasher